

KLEIN BOTTLE SETS OF DOUBLING $3|S| - 3$ REFUTE A TORSION-FREE SMALL-DOUBLING CONJECTURE OF MOHAN AND NEETU AS STATED

ABSTRACT. Let S be a finite subset of a group G and $S^2 = \{s_1 s_2 : s_1, s_2 \in S\}$. Conjecture 6.3 of Mohan and Neetu (arXiv:2607.11194v1) asserts that if G is torsion-free, $|S| \geq 3$, $e \in S$ and $|S^2| \leq 3|S| - 3$, then $\langle S \rangle$ is abelian. We exhibit counterexamples, at every admissible cardinality, inside the Klein bottle group $K = \mathbb{Z} \rtimes_{-1} \mathbb{Z} = \langle a, t \mid tat^{-1} = a^{-1} \rangle$: writing $a^m t^n = (m, n)$ and

$$S_k = \{(0, 0)\} \cup \{(i, 1) : 1 \leq i \leq k-1\} \quad (k \geq 3),$$

one has $|S_k| = k$, $e \in S_k$, $|S_k^2| = 3k - 3 = 3|S_k| - 3$ and $\langle S_k \rangle = K$, which is nonabelian. Moreover K carries a right-invariant total order under which $e = \min S_k$. Freiman's torsion-free conjecture at $|S^2| \leq 3|S| - 4$ is not contradicted; the family sits one unit above it.

1. THE CONJECTURE

Throughout, G is a group with identity e , $S \subseteq G$ is finite, and

$$S^2 = \{s_1 s_2 : s_1, s_2 \in S\}$$

is the product set over *ordered* pairs, so that $s^2 \in S^2$ for every $s \in S$ and $S \subseteq S^2$ as soon as $e \in S$. A set S is called *abelian* when $\langle S \rangle$ is an abelian subgroup of G . In these conventions, which are those of [4], Freiman, Herzog, Longobardi and Maj proved [3, Theorem 1.3] that if G is *orderable*, $|S| \geq 3$ and $|S^2| \leq 3|S| - 3$, then S is abelian — with no hypothesis $e \in S$. Mohan and Neetu ask whether the hypothesis “orderable” can be weakened to “torsion-free” at the cost of adding $e \in S$. Verbatim [4, Conjecture 6.3], quoted from the e-print arXiv:2607.11194v1 of 13 July 2026, its only version, where the statement stands in Section 6:

Conjecture 6.3 (Mohan and Neetu). Let S be a nonempty finite subset of a torsion-free group G with $|S| \geq 3$ and $e \in S$. If $|S^2| \leq 3|S| - 3$, then $\langle S \rangle$ is an abelian subgroup of G .

2. THE COUNTEREXAMPLE

Let K be the Klein bottle group,

$$K = \mathbb{Z} \rtimes_{-1} \mathbb{Z} = \langle a, t \mid tat^{-1} = a^{-1} \rangle = \text{BS}(1, -1),$$

2020 *Mathematics Subject Classification.* 11P70, 20F60, 20F65, 11B13.

Key words and phrases. small doubling, product set, torsion-free group, Klein bottle group, Baumslag-Solitar group, right-ordered group, counterexample.

in the coordinates $a^m t^n \leftrightarrow (m, n)$, $m, n \in \mathbb{Z}$. The multiplication and the inverse are

$$(1) \quad (m, n)(p, q) = (m + (-1)^n p, n + q), \quad (m, n)^{-1} = (-(-1)^n m, -n),$$

with identity $e = (0, 0)$.

Lemma 1. *K is torsion-free, and it is not abelian.*

Proof. By (1) the second coordinate is additive, so $(m, n)^r$ has second coordinate rn ; if $n \neq 0$ and $r \neq 0$ this is nonzero and $(m, n)^r \neq e$. If $n = 0$ then $(m, 0)^r = (rm, 0)$, which is $\neq e$ when $m \neq 0$ and $r \neq 0$. Hence no element other than e has finite order. Finally $(1, 0)(0, 1) = (1, 1)$ while $(0, 1)(1, 0) = (-1, 1)$, so K is nonabelian. $\square$

For an integer $k \geq 3$ put $I = \{1, 2, \dots, k-1\}$ and

$$(2) \quad S_k = \{(0, 0)\} \cup \{(i, 1) : i \in I\} = \{e, at, a^2t, \dots, a^{k-1}t\}.$$

Theorem 2. *For every integer $k \geq 3$ the set $S_k \subseteq K$ of (2) satisfies*

$$|S_k| = k, \quad e \in S_k, \quad S_k^2 = S_k \cup \{(d, 2) : |d| \leq k-2\},$$

$$|S_k^2| = 3k - 3 = 3|S_k| - 3,$$

while $\langle S_k \rangle = K$ is nonabelian. Since K is torsion-free by Lemma 1, Conjecture 6.3 of [4] is false as stated — and false for every admissible cardinality $|S| = k \geq 3$, not at a single value.

Proof. The k listed elements of (2) are pairwise distinct because their coordinates are, and $e = (0, 0) \in S_k$.

Split S_k^2 by the second coordinate, which by (1) adds and therefore separates the three blocks below.

- Second coordinate 0: the only product of two elements of S_k with second coordinate 0 is $e \cdot e = e$. One element.
- Second coordinate 1: the products are $e \cdot (i, 1) = (i, 1) = (i, 1) \cdot e$ for $i \in I$. Exactly the $k-1$ elements $\{(i, 1) : i \in I\}$, and nothing new.
- Second coordinate 2: for $i, j \in I$, $(i, 1)(j, 1) = (i + (-1)^1 j, 2) = (i - j, 2)$, so this block is $\{(d, 2) : d \in I - I\}$. As $I = \{1, \dots, k-1\}$ is an interval of integers, $I - I = \{-(k-2), \dots, k-2\}$ and $|I - I| = 2k-3$ exactly.

Hence $S_k^2 = S_k \cup \{(d, 2) : |d| \leq k-2\}$ and

$$|S_k^2| = 1 + (k-1) + (2k-3) = 3k-3 = 3|S_k| - 3,$$

so the antecedent of Conjecture 6.3 holds, with equality.

For the consequent, $(1, 1)(2, 1) = (-1, 2)$ and $(2, 1)(1, 1) = (1, 2)$, so S_k is not even commuting elementwise. Moreover $\langle S_k \rangle = K$: by (1) $(1, 1)^{-1} = (1, -1)$, and

$$(1, -1)(2, 1) = (1 - 2, 0) = (-1, 0) = a^{-1}, \quad (-1, 0)(1, 1) = (0, 1) = t,$$

so $\langle S_k \rangle$ contains a and t , hence equals K , which is nonabelian by Lemma 1. $\square$

The case $k = 3$. Here $S_3 = \{e, at, a^2t\} = \{(0, 0), (1, 1), (2, 1)\}$, and the nine ordered products are

$$\begin{array}{lll} e \cdot e = (0, 0), & e \cdot (1, 1) = (1, 1), & e \cdot (2, 1) = (2, 1), \\ (1, 1) \cdot e = (1, 1), & (2, 1) \cdot e = (2, 1), & (1, 1)(1, 1) = (0, 2), \\ (2, 1)(2, 1) = (0, 2), & (1, 1)(2, 1) = (-1, 2), & (2, 1)(1, 1) = (1, 2), \end{array}$$

with a single collision, $(at)^2 = (a^2t)^2 = t^2$. Thus

$$S_3^2 = \{(0, 0), (1, 1), (2, 1), (-1, 2), (0, 2), (1, 2)\}, \quad |S_3^2| = 6 \leq 3 \cdot 3 - 3 = 6.$$

Corollary 3. *Order K by*

$$(m, n) < (m', n') \iff n < n', \text{ or } n = n' \text{ and } m < m'.$$

This is a right-invariant total order on K , and $\min S_k = e$ for every $k \geq 3$.

Proof. Right multiplication by (p, q) sends (m, n) to $(m + (-1)^n p, n + q)$. If $n < n'$ then $n + q < n' + q$. If $n = n'$ then both first coordinates are shifted by the same integer $(-1)^n p$, so their order is preserved. Totality and transitivity are those of the lexicographic order on $\mathbb{Z}^2$. Every element of $S_k \setminus \{e\}$ has second coordinate $1 > 0$, so $e = \min S_k$. $\square$

Remark (scope). No minimality is claimed: nothing above asserts that $3|S| - 3$ is the least doubling attainable by an identity-containing finite set with nonabelian span in a torsion-free group, nor that $k = 3$ is the least cardinality in any other host group. Freiman's torsion-free conjecture concerns $|S^2| \leq 3|S| - 4$ [2, §1], and our family has $|S_k^2| = 3k - 3 > 3k - 4$, one unit above that threshold, so it is consistent with that conjecture. The host group is $\text{BS}(1, -1)$, which is the case excluded in [4, Theorem 2.11]; what is refuted is Conjecture 6.3 as printed, and nothing here bears on a version of it in which that exclusion is in force. No priority or novelty is claimed for the object: K together with a three-element identity-containing set of nonabelian span occurs in [2, §1] and in [1], and the companion paper [5] is closed access and was not available to us.

3. VERIFICATION

Every object the note claims is printed above: the group law (1), the family (2), the nine products of the $k = 3$ cell, the closed form for S_k^2 , the generating words for a^{-1} and t , and the order of Corollary 3. A reader therefore needs no code. A short program (`verify.py`, with its recorded output in `verify.output.txt`) accompanies the note anyway: it re-implements (1) in exact integer arithmetic, recomputes S_k^2 product by product for $3 \leq k \leq 30$ and compares it with the closed form as a set of coordinate pairs rather than by cardinality, re-derives the nine displayed products, and verifies right-invariance of the order together with $\min S_k = e$; it also runs controls of both polarities, so that neither of the two detectors it uses can be a constant. All 304 of its checks pass. It states in its own output what it does not cover: torsion-freeness is confirmed structurally and over a bounded box only, the

general statement being Lemma 1; no search for a smaller counterexample is performed; and no bibliographic claim is checked.

REFERENCES

- [1] A. Abdollahi and F. Jafari, *Cardinality of product sets in torsion-free groups and applications in group algebras*, J. Algebra Appl. **19** (2020), no. 5, 2050079; arXiv:1808.08708.
- [2] K. J. Böröczky, P. P. Pálffy, and O. Serra, *On the cardinality of sumsets in torsion-free groups*, Bull. London Math. Soc. **44** (2012), no. 5, 1034–1041. doi:10.1112/blms/bds032; arXiv:1009.6140.
- [3] G. A. Freiman, M. Herzog, P. Longobardi, and M. Maj, *Small doubling in ordered groups*, J. Aust. Math. Soc. **96** (2014), no. 3, 316–325. Theorem 1.3 is quoted as it is restated in [4, Theorem 2.3].
- [4] Mohan and Neetu, *On small doubling in right-ordered groups and Baumslag–Solitar groups – II*, arXiv:2607.11194v1, 2026. arXiv:2607.11194. Statement numbers are quoted from this e-print.
- [5] Mohan, Neetu, and Shankar, *On small doubling in right-ordered groups and Baumslag–Solitar groups*, Results Math. **81** (2026), no. 1, art. 12. doi:10.1007/s00025-025-02576-2. Closed access, no e-print.